Hence by rewriting the correction function as

$$f_{2,1,[\![\mathrm{OABAO}]\!]} = \frac{\beta}{24}\left(3\mathcal{L}_{\mathrm{A}}^{\dagger}|\nabla U|^2 - \frac{1}{3}\left(\mathcal{L}_{\mathrm{A}}^{\dagger}\right)^3 U - \frac{2}{\beta}\mathcal{L}_{\mathrm{A}}^{\dagger}\Delta_q U\right) + \mathcal{L}_{\mathrm{A}}^{\dagger}\nu - \mu,$$

we have

$$\bar{\pi}\mathcal{L}_H^{\dagger} f_{2,1,[\![\mathrm{OABAO}]\!]} = \frac{\beta}{24}\left(3\mathcal{M}|\nabla U|^2 - \frac{1}{3}\mathcal{N}U - \frac{2}{\beta}\mathcal{M}\Delta_q U\right) + \mathcal{M}\nu,$$

where we abbreviate

$$\mathcal{M}\psi(\boldsymbol{q}) = \bar{\pi}\mathcal{L}_H^{\dagger}\mathcal{L}_{\mathrm{A}}^{\dagger}\psi(\boldsymbol{q}) = \bar{\pi}\left((\mathcal{L}_{\mathrm{A}}^{\dagger})^2 + \mathcal{L}_{\mathrm{B}}^{\dagger}\mathcal{L}_{\mathrm{A}}^{\dagger}\right)\psi(\boldsymbol{q}) = \frac{1}{\beta}\Delta_q\psi(\boldsymbol{q}) - \nabla U\cdot\nabla_q\psi(q),$$

$$\mathcal{N}\psi(\boldsymbol{q}) = \bar{\pi}\mathcal{L}_H^{\dagger}(\mathcal{L}_{\mathrm{A}}^{\dagger})^3\psi(\boldsymbol{q}) = \bar{\pi}\left((\mathcal{L}_{\mathrm{A}}^{\dagger})^4 + \mathcal{L}_{\mathrm{B}}^{\dagger}(\mathcal{L}_{\mathrm{A}}^{\dagger})^3\right)\psi(\boldsymbol{q}) = \frac{3}{\beta}\mathcal{M}\left(\Delta_q\psi(\boldsymbol{q})\right).$$

Thus we have

$$\bar{\pi}\mathcal{L}_H^{\dagger}\rho_\beta f_{2,1,[\![\mathrm{OABAO}]\!]} = \frac{\beta}{8}\mathcal{M}\left(|\nabla U|^2 - \frac{1}{\beta}\Delta_q U\right) + \mathcal{M}\nu,$$

and hence by (7.36) we finally obtain

$$\nu(\boldsymbol{q}) = \frac{1}{8}\left(\Delta_q U - \beta|\nabla U|^2\right),$$

where it is easily verified that $\langle\nu\rangle = 0$. This gives us our leading order solution

$$f_{2,0,[\![\mathrm{OABAO}]\!]} = \frac{1}{8}\left(\Delta_q U - \beta|\nabla U|^2\right).$$

Applying the same procedure for the $[\![\mathrm{OBABO}]\!]$ scheme yields

$$f_{2,0,[\![\mathrm{OBABO}]\!]} = \frac{1}{8}\left(\beta|\nabla U|^2 - 2\Delta_q U + c\right),$$

where $c = \langle\Delta U\rangle$ is a constant chosen to ensure zero average. We can relate these schemes to their conjugate schemes with centralized OU update via Lemma 7.1 and Eq. (7.30), to give

$$f_{2,0,[\![\mathrm{ABOBA}]\!]} = \frac{1}{8}\left(\beta\boldsymbol{p}^T\nabla^2 U(\boldsymbol{q})\boldsymbol{p} - 2\Delta_q U + c\right),$$

$$f_{2,0,[\![\mathrm{BAOAB}]\!]} = \frac{1}{8}\left(\Delta U - \beta\boldsymbol{p}^T\nabla^2 U(\boldsymbol{q})\boldsymbol{p}\right).$$

Owing to our ansatz, we would expect that the computed average of some observable has behavior like

$$\langle \vartheta \rangle_h = \langle \vartheta \rangle + h^2 \langle \vartheta f_{2,0} \rangle + \mathcal{O}(\varepsilon h^2 + h^4).$$

At sufficiently small values of ε (and hence, large values of the friction coefficient γ) we would expect that the $\mathcal{O}(h^2)$ remainder will not be visible to us in our experiments, and we will be able to correct our average to $\mathcal{O}(h^4)$. The symmetric discretization schemes we have considered here will all be consistent in the overdamped ($\gamma \to \infty$) limit, and we would expect the correction function for the corresponding limiting schemes to be given by $f_{2,0}$.

7.9.3 *Superconvergence of the* ⟦BAOAB⟧ *Scheme*

In the regime of sufficiently large γ and small h, our doubly-asymptotic ansatz for the leading order correction to a function $\vartheta(q,p)$ is

$$\langle \vartheta \rangle = \langle \vartheta \rangle_h - h^2 \langle \vartheta f_{2,0} \rangle + O\left(\frac{h^2}{\gamma} + h^4\right), \tag{7.37}$$

where $f_{2,0}$ is method-dependent. The ⟦BAOAB⟧ scheme has leading order correction

$$f_{2,0,[\![\mathrm{BAOAB}]\!]}(\boldsymbol{q},\boldsymbol{p}) = \frac{1}{8}\left(\Delta U - \beta \boldsymbol{p}^T \nabla^2 U(\boldsymbol{q})\boldsymbol{p}\right),$$

which has the interesting property that its marginal average with respect to momentum is zero:

$$\bar{\pi} f_{2,0,[\![\mathrm{BAOAB}]\!]} = \int_{\mathcal{P}} f_{2,0,[\![\mathrm{BAOAB}]\!]}(\boldsymbol{q},\boldsymbol{p}) \rho_\beta(\boldsymbol{q},\boldsymbol{p}) \mathrm{d}\omega_p = 0.$$

This is a very favorable quality for configurational sampling, where we consider averages of functions solely of position, as we can split up the integral such that

$$\begin{aligned}
\langle \vartheta(\boldsymbol{q}) f_{2,0,[\![\mathrm{BAOAB}]\!]} \rangle &= \int_{\mathcal{D}} \vartheta(\boldsymbol{q}) f_{2,0,[\![\mathrm{BAOAB}]\!]} \rho_\beta \mathrm{d}\omega, \\
&= \int_{\mathcal{Q}} \vartheta(\boldsymbol{q}) \int_{\mathcal{P}} f_{2,0,[\![\mathrm{BAOAB}]\!]} \rho_\beta \mathrm{d}\omega_p \mathrm{d}\omega_q, \\
&= 0.
\end{aligned}$$